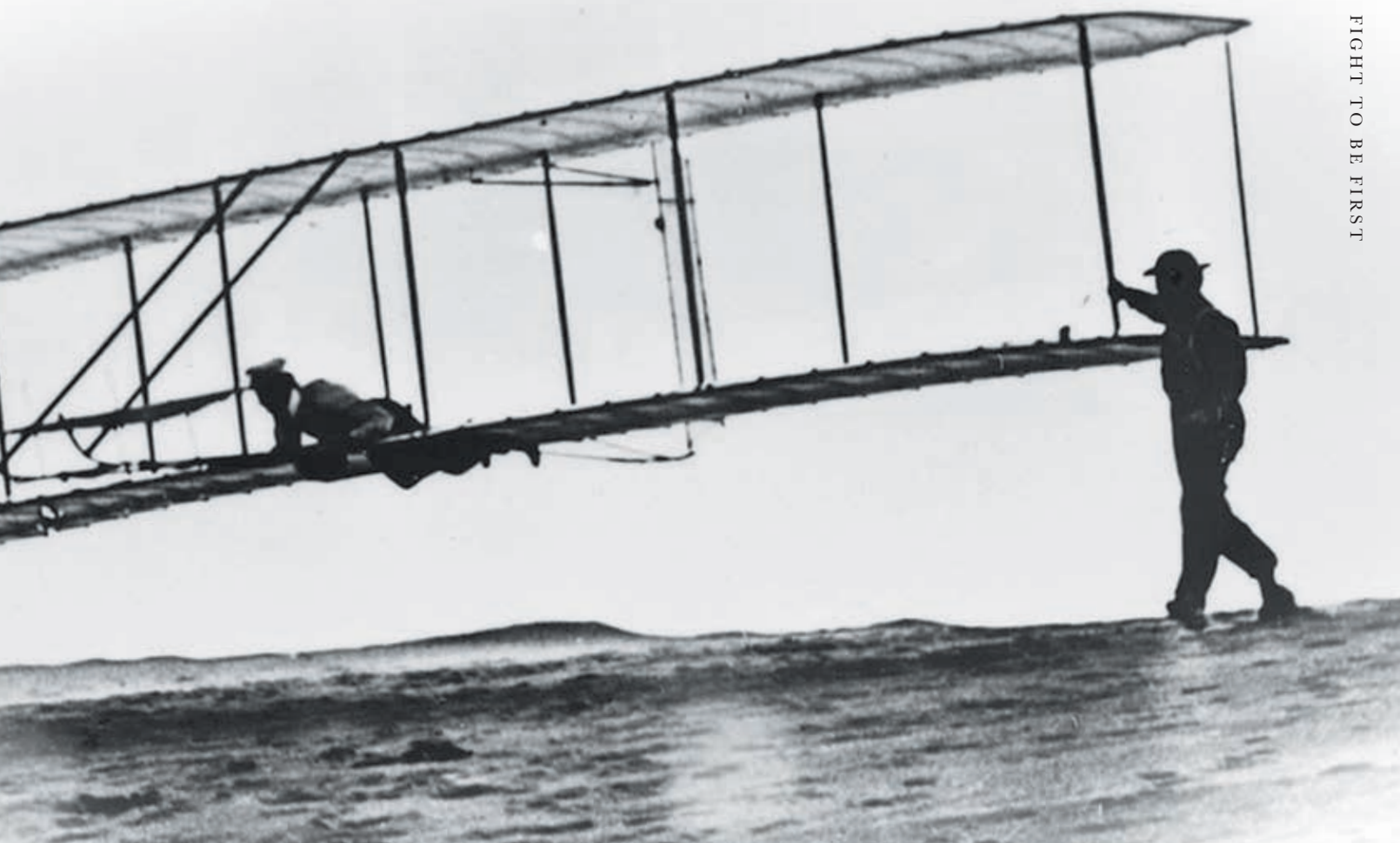


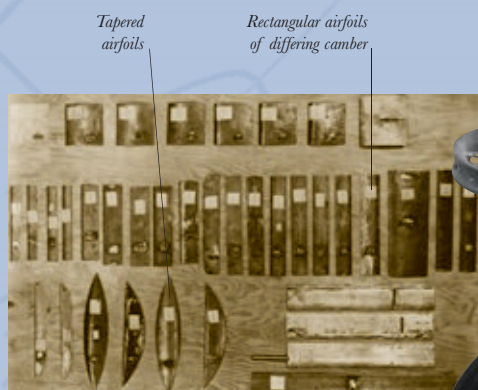


## THE WRIGHTS' RIGOROUS EXPERIMENTS

IN THE WINTER OF 1901–02, the Wright brothers carried out a remarkable set of experiments that overturned accepted wisdom on wing design. Their goal was to test a wide range of potential wing shapes in order to establish their aerodynamic characteristics.

First, they conducted fairly rough-and-ready experiments on a bicycle. As one of the brothers cycled along to create an airflow, he then adjusted the angle of the airfoil until it balanced the wind pressure on the flat plate (see far right).

Although the bicycle experiments showed that previously published figures (by Lilienthal and others) were wrong, they lacked precision. The key experimental challenge was to create a perfectly controlled airflow and an exact record of the resulting performance. For this the Wrights built a wind tunnel, in which they tested airfoils for two months under controlled conditions. This gave them a highly accurate series of figures that they were then able to apply to wing design.



### AIRFOIL TESTING

*The Wright brothers made about 200 variously shaped airfoils out of sheet metal (above) and set out to record the performance of each in a wind tunnel. The bicycle-mounted device on the right has two surfaces attached to it: an experimental airfoil and a flat metal plate for comparison.*

